Zephyr Sensor Fusion with MicroBlaze V on Kria KR260 and OpenAMP

Detail Design Guide2- Zephyr Device Tree Creation and Building Zephyr App

A. Jumpstart on Zephyr Device Tree

One can follow this readme document from the repository to modify the pl.dtsi and system_top.dts and generating mbv32.dtsusing **`LOPPER_DTC_FLAGS`** command:

README.md#2---zephyr-device-tree

Additionally, those steps are also mentioned below,

Creating Zephyr Device Tree

First activate the zephyr environment:

```
logictronix06@logictronix06:~/zephyrproject$ source ~/zephyrproject/.venv/bin/activate
(.venv) logictronix06@logictronix06:~/zephyrproject$ cd zephyr
```

One can run the looper command to generate the Zephyr device tree:

```
LOPPER_DTC_FLAGS="-b 0 -@" west lopper-command -p microblaze_riscv_0 -
s <path_of_device_tree>/zephyr/zephyr/system-top.dts -w .
```

Below is the screenshot covering the looper command output:

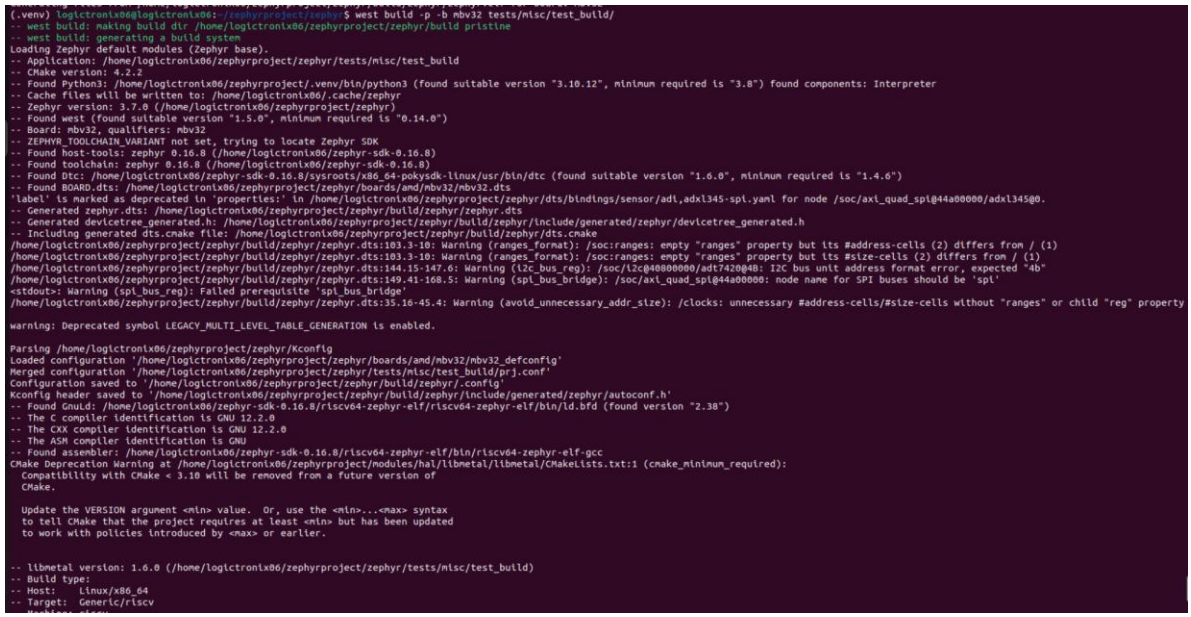
```
(.venv) logictronix06@logictronix06:~/zephyrproject/zephyr$ LOPPER_DTC_FLAGS="-b 0 -@" west lopper-command -p microblaze_riscv_0 -s /home/logictronix06/Videos/25_26_prj/Zephyr-Op
enAMP-MBlze-KR260/device_tree/zephyr/system-top.dts -w ~/zephyrproject/zephyr
{'__dbg__': 0,
 '__modified__': False,
 '__pstate__': 'resolved',
 'abs_path': '/cpus_microblaze_riscv@0/cpu@0/xlnx,data-size',
 'binary': False,
 'name': 'xlnx,data-size',
 'node': <lopper.tree.LopperNode object at 0x757a3754a750>,
 'number': -1,
 'pclass': <class 'list'>,
 'phandle_resolution': True,
 'ptype': <LopperFmt.UINT32: 8>,
 'string_val': 'xlnx,data-size = <0x20>;',
 'value': [32]}
(.venv) logictronix06@logictronix06:~/zephyrproject/zephyr$
```

After obtaining the mbv32.dts, now one can add the GPIO node and SPI node on the device tree as mentioned at the readme linked above.

B. Building Zephyr Application

One can run following command to build the Zephyr application:

```
west build -p -b mbv32 tests/misc/test_build/
```



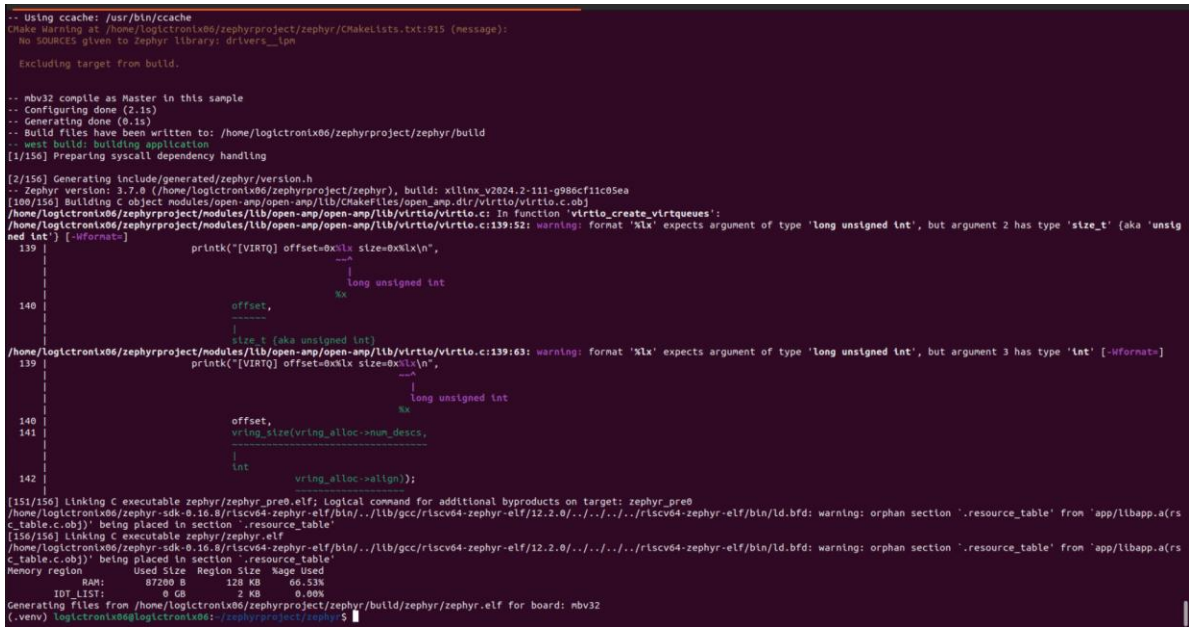
```
(.venv) logictronix06@logictronix06:~/zephyrproject/zephyr$ west build -p -b mbv32 tests/misc/test_build/
-- west build: making build dir /home/logictronix06/zephyrproject/zephyr/build pristine
-- west build: generating a build system
Loading Zephyr default modules (Zephyr base).
-- Application: /home/logictronix06/zephyrproject/zephyr/tests/misc/test_build
-- CMake version: 4.2.2
-- Found Python3: /home/logictronix06/zephyrproject/.venv/bin/python3 (found suitable version "3.10.12", minimum required is "3.8") found components: Interpreter
-- Cache files will be written to: /home/logictronix06/.cache/zephyr
-- Zephyr version: 3.7.0 (/home/logictronix06/zephyrproject/zephyr)
-- Found west (found suitable version "1.5.0", minimum required is "0.14.0")
-- Board: mbv32, qualifiers: mbv32
-- ZEPHYR_TOOLCHAIN_VARIANT not set, trying to locate Zephyr SDK
-- Found host-tools: zephyr 0.16.8 (/home/logictronix06/zephyr-sdk-0.16.8)
-- Found toolchain: zephyr 0.16.8 (/home/logictronix06/zephyr-sdk-0.16.8)
-- Found Dtc: /home/logictronix06/zephyr-sdk-0.16.8/sysroots/x86_64-pokysdk-linux/usr/bin/dtc (found suitable version "1.6.0", minimum required is "1.4.6")
-- Found BOARD.dts: /home/logictronix06/zephyrproject/zephyr/boards/amd/mbv32/mbv32.dts
'label' is marked as deprecated in 'properties:' in /home/logictronix06/zephyrproject/zephyr/dts/bindings/sensor/adi,adxl345-spi.yaml for node /soc/axi_quad_spi@44a00000/adxl345@0.
-- Generated zephyr.dts: /home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.dts
-- Generated devicetree_generated.h: /home/logictronix06/zephyrproject/zephyr/build/zephyr/include/generated/zephyr/devicetree_generated.h
-- Including generated dts.cmake file: /home/logictronix06/zephyrproject/zephyr/build/zephyr/dts.cmake
/home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.dts:103.3-10: Warning (ranges_format): /soc:ranges: empty "ranges" property but its #address-cells (2) differs from / (1)
/home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.dts:103.3-10: Warning (ranges_format): /soc:ranges: empty "ranges" property but its #size-cells (2) differs from / (1)
/home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.dts:144.15-147.6: Warning (i2c_bus_reg): /soc/i2c@40800000/adt7420@48: I2C bus unit address format error, expected "4b"
/home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.dts:149.41-168.5: Warning (spi_bus_bridge): /soc/axi_quad_spi@44a00000: node name for SPI buses should be 'spi'
<stdout>: Warning (spi_bus_reg): Failed prerequisite 'spi_bus_bridge'
/home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.dts:35.16-45.4: Warning (avoid_unnecessary_addr_size): /clocks: unnecessary #address-cells/#size-cells without "ranges" or child "reg" property

warning: Deprecated symbol LEGACY_MULTI_LEVEL_TABLE_GENERATION is enabled.

Parsing /home/logictronix06/zephyrproject/zephyr/Kconfig
Loaded configuration '/home/logictronix06/zephyrproject/zephyr/boards/amd/mbv32/mbv32_defconfig'
Merged configuration '/home/logictronix06/zephyrproject/zephyr/tests/misc/test_build/prj.conf'
Configuration saved to '/home/logictronix06/zephyrproject/zephyr/build/zephyr/.config'
Kconfig header saved to '/home/logictronix06/zephyrproject/zephyr/build/zephyr/include/generated/zephyr/autoconf.h'
-- Found GnuLd: /home/logictronix06/zephyr-sdk-0.16.8/riscv64-zephyr-elf/riscv64-zephyr-elf/bin/ld.bfd (found version "2.38")
-- The C compiler identification is GNU 12.2.0
-- The CXX compiler identification is GNU 12.2.0
-- The ASM compiler identification is GNU
-- Found assembler: /home/logictronix06/zephyr-sdk-0.16.8/riscv64-zephyr-elf/bin/riscv64-zephyr-elf-gcc
CMake Deprecation Warning at /home/logictronix06/zephyrproject/modules/hal/libmetal/libmetal/CMakeLists.txt:1 (cmake_minimum_required):
  Compatibility with CMake < 3.10 will be removed from a future version of
  CMake.

  Update the VERSION argument <min> value.  Or, use the <min>...<max> syntax
  to tell CMake that the project requires at least <min> but has been updated
  to work with policies introduced by <max> or earlier.


-- libmetal version: 1.6.0 (/home/logictronix06/zephyrproject/zephyr/tests/misc/test_build)
-- Build type:
-- Host:    Linux/x86_64
-- Target:  Generic/riscv
```



```
-- Using ccache: /usr/bin/ccache
CMake Warning at /home/logictronix06/zephyrproject/zephyr/CMakeLists.txt:915 (message):
  No SOURCES given to Zephyr library: drivers__lpm

  Excluding target from build.


-- mbv32 compile as Master in this sample
-- Configuring done (2.1s)
-- Generating done (0.1s)
-- Build files have been written to: /home/logictronix06/zephyrproject/zephyr/build
-- west build: building application
[1/156] Preparing syscall dependency handling

[2/156] Generating include/generated/zephyr/version.h
-- Zephyr version: 3.7.0 (/home/logictronix06/zephyrproject/zephyr), build: xilinx_v2024.2-111-g986cf11c05ea
[100/156] Building C object modules/open-amp/open-amp/lib/CMakeFiles/open_amp.dir/virtio/virtio.c.obj
/home/logictronix06/zephyrproject/modules/lib/open-amp/open-amp/lib/virtio/virtio.c: In function 'virtio_create_virtqueues':
/home/logictronix06/zephyrproject/modules/lib/open-amp/open-amp/lib/virtio/virtio.c:139:52: warning: format '%lx' expects argument of type 'long unsigned int', but argument 2 has type 'size_t' (aka 'unsig
ned int') [-Wformat=]
  139 |                         printk("[VIRTQ] offset=0x%lx size=0x%lx\n",
      |                                                   ~~^
      |                                                     |
      |                                                     long unsigned int
      |                                                   %x
  140 |                                 offset,
      |                                 ~~~~~~
      |                                 |
      |                                 size_t (aka unsigned int)
/home/logictronix06/zephyrproject/modules/lib/open-amp/open-amp/lib/virtio/virtio.c:139:63: warning: format '%lx' expects argument of type 'long unsigned int', but argument 3 has type 'int' [-Wformat=]
  139 |                         printk("[VIRTQ] offset=0x%lx size=0x%lx\n",
      |                                                         ~~^
      |                                                           |
      |                                                           long unsigned int
      |                                                         %x
  140 |                                 offset,
  141 |                                 vring_size(vring_alloc->num_descs,
      |                                 ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
      |                                 |
      |                                 int
  142 |                                         vring_alloc->align));
      |                                         ~~~~~~~~~~~~~~~~~~~~
[151/156] Linking C executable zephyr/zephyr_pre0.elf; Logical command for additional byproducts on target: zephyr_pre0
/home/logictronix06/zephyr-sdk-0.16.8/riscv64-zephyr-elf/bin/../lib/gcc/riscv64-zephyr-elf/12.2.0/../../../../riscv64-zephyr-elf/bin/ld.bfd: warning: orphan section `.resource_table' from `app/libapp.a(rs
c_table.c.obj)' being placed in section `.resource_table'
[156/156] Linking C executable zephyr/zephyr.elf
/home/logictronix06/zephyr-sdk-0.16.8/riscv64-zephyr-elf/bin/../lib/gcc/riscv64-zephyr-elf/12.2.0/../../../../riscv64-zephyr-elf/bin/ld.bfd: warning: orphan section `.resource_table' from `app/libapp.a(rs
c_table.c.obj)' being placed in section `.resource_table'
Memory region         Used Size  Region Size  %age Used
             RAM:       87200 B       128 KB     66.53%
        IDT_LIST:          0 GB         2 KB      0.00%
Generating files from /home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.elf for board: mbv32
(.venv) logictronix06@logictronix06:~/zephyrproject/zephyr$
```

At the end of Zephyr west build, one will get the ***zephyr.elf*** file, this is the application file which we will be using in upcoming step.

In general, we will merge this ***zephyr.elf*** with the BIT file and create a single BIT only, so that we can load the BIT by renaming it to BIN at Kria Firmware load with ***xmutil***.

Revision History

S.N	Revision Number	Revision Details	Author
1.	v1.0	Initial Revision	Krishna Gaihre and Sanam Shakya